SARATH RADHAKRISHNAN

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PROFESSIONAL SUMMARY

Researcher with expertise in computational fluid dynamics, turbulence-resolving simulations, non-linear flow physics, and scientific computing. Experienced in high-fidelity numerical methods, surrogate modeling, and machine learning approaches for engineering applications, including the development of surrogate and reduced-order models of complex systems. Background includes aerospace propulsion systems, large-scale CFD workflows, and multi-physics computational research. Interested in non-linear mechanics, fluid–structure interactions, scientific machine learning, and physics-informed modeling of complex dynamical systems.

WORK EXPERIENCE

Barcelona Supercomputing Center

2024 - 2025 | Barcelona, Spain

Recognized Researcher

- Contributed to the development of reduced-order and surrogate-modelling approaches for CFD applications
- Collaborated with interdisciplinary and international research teams across CFD and scientific-computing projects
- Supported graduate students in numerical modelling, data analysis, and scientific problem-solving

Barcelona Supercomputing Center

2018 - 2024 | Barcelona, Spain

First Stage Researcher

- Research focus: data-driven non-equilibrium wall modelling for Large-Eddy Simulation (LES) of turbulent wall-bounded flows
- Combined turbulence modelling, numerical simulation, and machine-learning methods
- Conducted high-fidelity wall-modeled large eddy simulations of complex three-dimensional flows involving strong anisotropy, separation, and non-linear turbulent interactions such as NASA Junction flow, NASA Hump, Neutral Atmospheric Boundary Layer flow over urban areas, etc.
- Developed machine-learning-assisted wall models for LES to represent unresolved near-wall turbulence in wall-bounded turbulent flows
- Analysed large numerical datasets using Python-based scientific computing and data-analysis workflows
- Performed HPC-based CFD workflows using parallel computing environments and large-scale simulation datasets
- Developed reproducible numerical-analysis and post-processing workflows for simulation data interpretation
- Contributed to technical publications, conference papers, and presentations in CFD and HPC-related research areas

Indian Space Research Organization (ISRO) Thiruvananthapuram, India | 2008 - 2015

System Engineer

- Worked on aerospace-related engineering projects within the Air Breathing Propulsion Project in a mission-oriented environment
- Contributed to engineering-analysis activities involving fluid mechanics, thermal-fluid systems, heat transfer, and structural mechanics
- Supported engineering activities related to propulsion-system-oriented thermo-fluid problems and associated engineering workflows
- Worked in interdisciplinary engineering teams operating under strict quality, reliability, and documentation standards
- Participated in technical problem-solving and engineering-analysis tasks for aerospace applications
- Mentored graduate engineering trainees in engineering fundamentals and analytical problem-solving

EDUCATION

Universitat Politecnica de Catalunia *2018 - 2024 | Barcelona, Spain*

PhD

Thesis: Non-Equilibrium Wall-modeling in Large Eddy Simulation of High-speed Transitional Flows

Advisors: Oriol Lehmkuhl and Daniel Mira

Ecole Centrale de Nantes *2015 - 2017 | France*

Master of Engineering in Computational Mechanics

- Advanced coursework in continuum mechanics, numerical methods, finite element methods, and computational modelling.
- Master's thesis focused on constitutive and finite-element modelling of biological tissues
- Performed finite-element simulation of uterine synchronization using ANSYS Parametric Design Language (APDL); thesis sponsored by ANSYS

National Institute of Technology Calicut *2004 - 2008 | India*

Bachelor of Technology in Mechanical Engineering

CERTIFICATIONS

Deep Learning Specialization, DeepLearning.AI 2021

Fundamentals of Parallelism on Intel Architecture, Intel 2021

SKILLS

CFD, Numerical Modeling & Scientific Computing: Computational Fluid Dynamics (CFD), Large Eddy Simulation (LES), Turbulence Modeling and Wall Modeling, Numerical Methods & Scientific Computing, Reduced Order and Surrogate Modeling, Data-driven and AI/

ML-assisted Modeling, Scientific workflow automation and post-processing , CFD data analysis and visualisation, verification, validation, and numerical-analysis awareness, GMSH, ANSA
Programming & Automation: Python (NumPy, SciPy, Pandas, Matplotlib) , Fortran, C/C++, Linux-based scientific computing workflows Scientific scripting and data processing, Matlab
High-Performance Computing: HPC environments and parallel workflows, Large-scale simulation data processing, Numerical workflow optimisation, Experience working with supercomputing systems
Engineering & Simulation Software: ANSYS (APDL & Mechanical), Abaqus (exposure) , Finite element-based in-house code development

PROJECTS

APPWIND

- Development of high resolution digital twins for fast prediction of air pollutants distribution and the odour impact in cities based on the application of AI to CFD models

HifiTurb

- Development of high-fidelity LES/DNS data for innovative turbulence models

vCity

- A human centric platform for urban digital twins

Air Breathing Propulsion Project

- Development of SCRAMJET

PUBLICATIONS & TECHNICAL CONTRIBUTIONS

Radhakrishnan, S., Calafell, J., Miró, A., Font, B., & Lehmkuhl, O. “Data-driven wall modeling for LES involving non-equilibrium boundary layer effects.” International Journal of Numerical Methods for Heat and Fluid Flow (Q1 Journal).

Radhakrishnan, S., Gyamfi, L.A., Miró, A., Font, B., Calafell, J., & Lehmkuhl, O. “A data-driven wall-shear stress model for LES using gradient boosted decision trees.” International Conference on High Performance Computing.

Owen, H., Radhakrishnan, S., Gomez, S., & Lehmkuhl, O. “A Finite-element Viscosity Based Non-slip Wall Modelling Strategy for Industrial Turbulent Flows.” ERCOFTAC Bulletin.

CONFERENCE PRESENTATIONS

International Supercomputing Conference (Online), 2021

Barcelona Supercomputing Center PhD Symposium, Barcelona, Spain, 2021

ParCFD Symposium, Alba, Italy, 2022
2nd Spanish Fluid Mechanics Conference, Barcelona, Spain, 2023

MENTORING & COLLABORATION

Supported and supervised graduate students during doctoral research activities
Guided students in numerical modelling, scientific computing, CFD data analysis, and engineering problem-solving
Experience collaborating in interdisciplinary and international research teams
Experience communicating technical findings through publications, presentations, and technical discussions

LANGUAGES

English (Fluent), Malayalam (Native), Spanish (Basic), Tamil (Fluent), Hindi (Basic)